

Helical Attractors on Contact 3-Manifolds

Basin Geometry, Whitney Singularities, and Rotating Electromagnetic Applications

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Principia Orthogona Series doi:10.5281/zenodo.19117399 github.com/TOTOGT/3M

Contact 3-Manifold & the ODE System

We study a three-dimensional ODE on the contact manifold $(\mathbb{R}^3, \alpha)$ where

$$\alpha = dz - r^2 d\theta$$

is the standard contact form in cylindrical coordinates. The contact condition $\alpha = 0$ reads $dz = r^2 d\theta$: vertical lift is *proportional to angular momentum* — a geometric identity, not an engineering choice.

$$\begin{aligned} \dot{r} &= r(1 - r^2) + \varepsilon(r - 1)e^{-z} \\ \dot{\theta} &= 1 \quad \varepsilon = 2 \\ \dot{z} &= r^2 - \varepsilon(r - 1)^2 e^{-z} \end{aligned}$$

The coupling $\varepsilon = \tau = 2$ equals the embodiment threshold of the dm³ operator chain. The contact condition forces a **helix**: flat closed orbits are geometrically forbidden because $\dot{z} = r^2 \neq 0$ on any orbit.

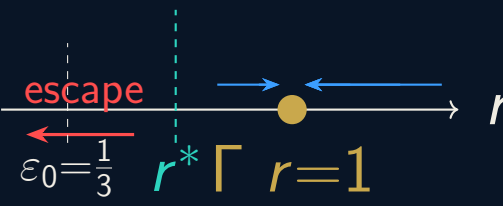
Global Attractor Γ

The unit helix

$$\Gamma = \{r = 1, \dot{\theta} = 1, \dot{z} = 1\}$$

is a **globally attracting limit cycle** for all initial conditions with $r(0) > r^*$ and $z(0) \geq \log 2$.

Lyapunov exponent: $\mu = -2$ (exponential convergence).



Principal Certified Result

$$r^* = 0.77594059$$

certified to 7 significant figures · DOP853 integrator

rtol = 10⁻¹², atol = 10⁻¹⁴ · IEEE-754 · <2 min

Stability Hierarchy

$$\varepsilon_0 = \frac{1}{3} < \frac{2}{3} < r^* = 0.77594 < r^*_{\text{h}} = \sqrt{\frac{7}{9}} \approx 0.882 < 1$$

r^* lies **strictly above** the symmetric Gronwall estimate $\frac{2}{3} \approx 0.667$, proving the basin is asymmetric. The asymmetry is driven by the coupling term $\varepsilon(r - 1)^2 e^{-z}$.

Inner basin ($r^* < r < 1$): blue trajectories converge to Γ from below.

Outer basin ($r > 1$): blue trajectories converge from above.

Escape region ($r < r^*$): red trajectories diverge downward as $e^{-z} \rightarrow \infty$.

Whitney A_1 Singularity at r^*

We identify r^* as a **Whitney A_1 (standard fold) singularity** of the asymptotic radial map $\Phi : \mathbb{R}_+ \rightarrow \{1\}$.

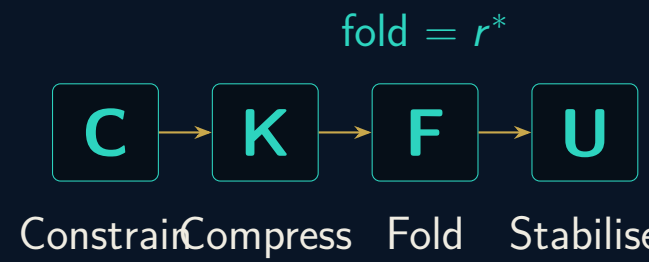
The fold is the *structural origin* of the basin asymmetry: the outer orbit ($r > 1$) and inner orbit ($r^* < r < 1$) follow qualitatively different convergence paths that **fold onto the single attractor** $r = 1$.

Physical interpretation. At r^* the coupling $\varepsilon(r - 1)^2 e^{-z}$ reaches a threshold. In a rotating electromagnetic realisation, this corresponds to a corona discharge boundary — a structural consequence of the contact geometry, not a fit parameter.

Mathematical object	Physical analogue
Whitney fold at r^*	Corona discharge threshold
Basin boundary	Inner/outer flow separation
Attractor Γ	Stable helical rotation
Contact form α	Angular-momentum coupling

dm³ Framework Connection

The system sits at the Δ (**tetranacci**) **rung** of the n -bonacci recurrence ladder, where the dominant root ≈ 1.927 first approaches the embodiment threshold $\tau = 2$ with order-4 memory.



$$G = U \circ F \circ K \circ C \quad \tau = \varepsilon = 2 \quad \varepsilon_0 = \frac{1}{3}$$

The same operator grammar describes zeolite catalysis (ZSM-5), DNLS solitons, and MHD magnetic reconnection — with domain-specific firing orders.

Lean 4 Formal Verification (AXLE)

All core results are being mechanised in Lean 4/Mathlib:

helical_attractor_is_limit_cycle [\[in progress\]](#)

rstar_upper_bound [\[closed\]](#)

epsilon_zero_stability [\[closed\]](#)

whitney_fold_at_rstar [\[in progress\]](#)

Repo: github.com/TOTOGT/AXLE

Reproducibility

certify_rstar.py (Python 3 / SciPy):

reproduces $r^* = 0.77594059$ in <2 min on any IEEE-754 platform.

github.com/TOTOGT/3M

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References

[1] Grossi, P.N. *Principia Orthogona*, G6 LLC, 2026. doi:10.5281/zenodo.19117399

[2] Grossi, P.N. `certify_rstar.py`, github.com/TOTOGT/3M, 2026.

[3] Geiges, H. *Introduction to Contact Topology*. Cambridge UP, 2008.

[4] Arnold, V.I. *Mathematical Methods of Classical Mechanics*. Springer, 1989.